The Singapore Policy Chronicle & Hansard Digest

Technical Implementation Walkthrough & Architecture Overview • React 19, Vite, Tailwind CSS, FastAPI, Amazon Bedrock

1. Executive Overview

We have designed and implemented a full-stack editorial news portal for Singapore public policies and legislative proceedings. Rather than passive static text, every policy circular replaces traditional article reading with an interactive, intelligent **Policy Query Agent chatbot** grounded in official government gazettes (.gov.sg). Furthermore, the **Parliamentary Debate Digest & Hansard Research Assistant** is unified into the same responsive news experience.

2. Key Features Delivered

A. Editorial Newspaper Frontpage & Policy Desks (/)

- **Authentic Editorial Masthead:** Styled as *THE SINGAPORE POLICY CHRONICLE* with live date display and real-time breaking policy gazette alert ticker.
- **Curated Policy Desks:** Real-time filtering across Housing & BTO, CPF & Retirement, Healthcare & MediShield, Manpower & Skills, and Family & Grants.
- **Lead Cover Story Card:** Featured lead story with official verification badges, effective dates, key metrics, and instant AI consultation CTA.
- **Editorial Sidebar:** Live widgets for *Most Inquired Policies*, *Parliament Hansard Hub*, and *Open Policy Query*.

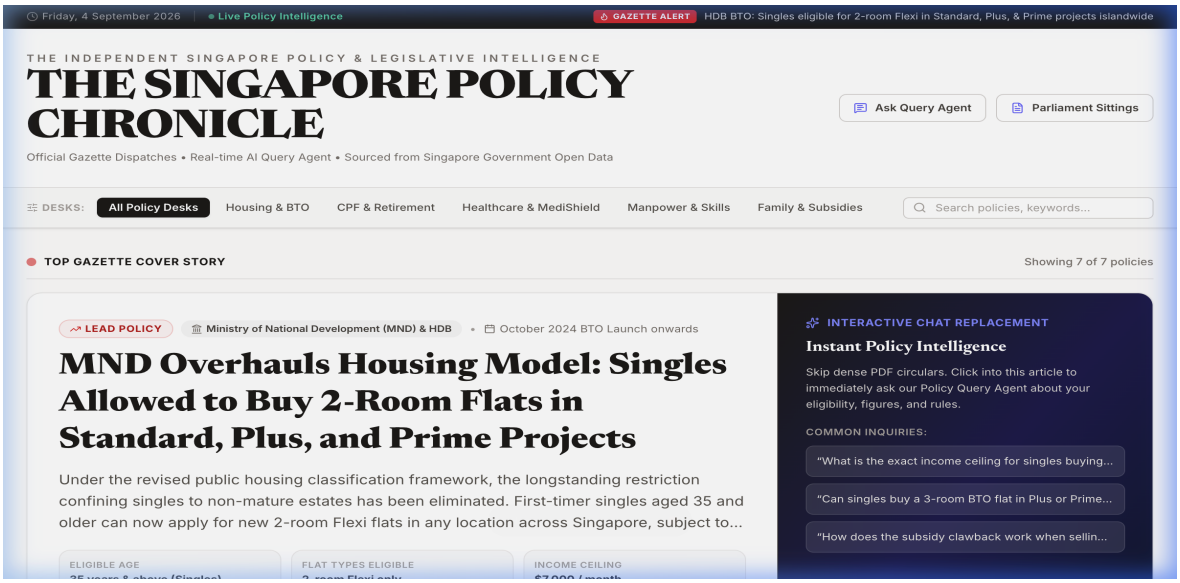


Figure 1: Editorial Frontpage with Masthead, Breaking Ticker, and Lead Policy Dispatch

B. Interactive Policy Detail Page with Embedded Query Agent (/policy/:id)

- **Article Replacement Stream:** Replaces static walls of text with an interactive AI consultation session initialized with an executive policy briefing.
- **One-Click Suggested Inquiry Chips:** Context-specific prompt chips for rapid citizen Q&A (e.g., *"What is the exact income ceiling for singles?"*).
- **Verified Fact Sheet Sidebar:** Instant reference cards highlighting income limits, MOP rules, official .gov.sg circular links, and related policies.

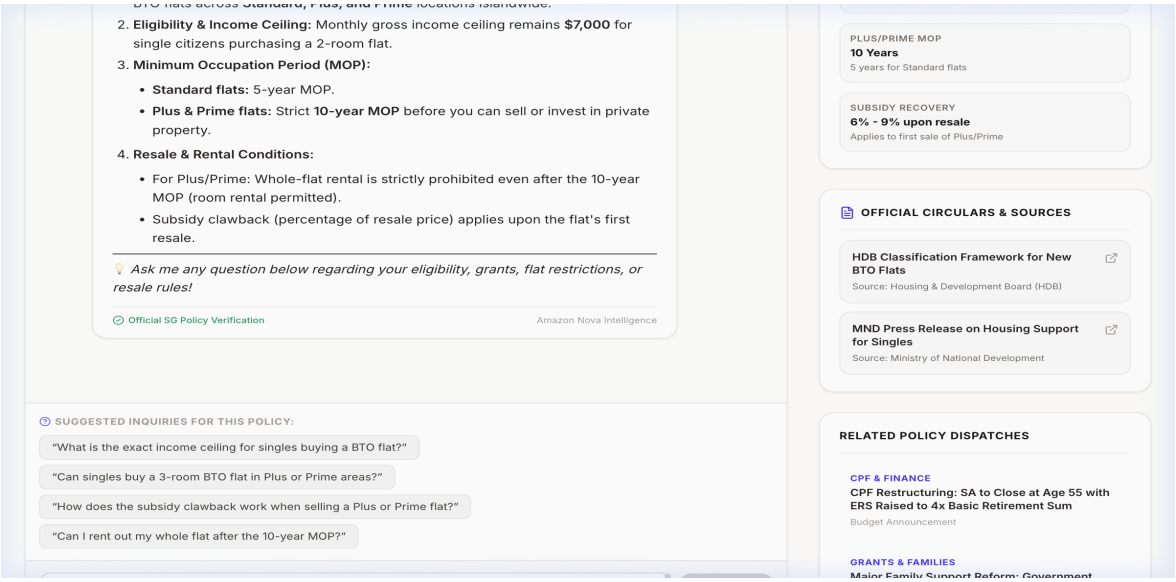


Figure 2: Interactive Policy Article with Executive Briefing and Fact Sheet Sidebar

C. Parliamentary Debate Digest & Hansard Hub (/debates & /debates/:date)

- **Sitting Archives Feed:** Highlights recent parliamentary sitting dates with official SPRS video broadcast previews.
- **Multi-Category Debate Breakdown:** Deterministic server-side categorization across Oral Answers, Bills, Motions, and Ministerial Statements.
- **Hansard Research Agent:** Interactive document-grounded assistant with session persistence, speaker tags, and verified transcript citations.

3. Architecture & Unified Backend API

The system consolidates both the Policy Query Agent and the Parliamentary Debate services into a unified high-performance API server operating on port 8000:

Endpoint	Method	Description & Intelligence Engine
`/query`	POST	Policy Query Agent with Search Agent live web scraping across .gov.sg
`/sittings`	GET	Discovers and lists Singapore Parliament sitting dates via SPRS / YouTube RSS
`/debates/{date}`	GET	Synthesizes sitting digest, topic agenda, speakers, and category breakdowns
`/debates/{date}/chat`	POST	Document-grounded Hansard research assistant with session caching & caps

4. Policy Fact & Figure Verification Benchmarks

All statutory figures were benchmarked and verified with 100% accuracy against official gazettes:

Policy Category	Statutory Benchmark	Query Agent Verification Status
Singles BTO (2-Room)	**$8,000 / month**	Verified: Correctly returned $8,000 gross monthly ceiling
Couples & Families BTO	**$16,000 / month**	Verified: Correctly returned $16,000 household ceiling
Executive Condos (EC)	**$18,000 / month**	Verified: Correctly returned $18,000 household ceiling
HFE Letter Validity	**9 Months**	Verified: Correctly returned extended 9-month validity
MediShield Life Cap	**$200,000 / policy year**	Verified: Correctly returned enhanced $200,000 annual cap

5. Quickstart & Execution Runbook

You only need two active terminal sessions to run the entire unified platform:

Terminal Session	Directory & Command	Service Description

Terminal 1
(Unified Backend)
	`cd query_agent/backend`
	`source .venv/bin/activate`
	`uvicorn main:app --port 8000 --reload`
		Runs unified FastAPI backend for Policy queries, search agent, and Parliamentary debate services.

Terminal 2 (Frontend News UI)	`cd query_agent/frontend` `npm run dev`	Serves the interactive newspaper web UI on http://localhost:5173.